

Zhixing You

Postdoctoral Researcher

Einstein Institute of Mathematics
The Hebrew University of Jerusalem, Edmond J. Safra Campus
✉ zhixing.you@mail.huji.ac.il
Google Scholar

Work Experience

- Oct 2025 – Present **Postdoctoral Researcher**, *Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Israel*, Advisor: Yair Hayut
- Jul 2022 – Sep 2025 **Postdoctoral Researcher**, *Department of Mathematics, Bar-Ilan University, Israel*, Advisor: Assaf Rinot

Education

- Sep 2016 – Jun 2022 **Ph.D. in Basic Mathematics**, *Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China*, Advisors: Joan Bagaria, Liuzhen Wu, Jiachen Yuan
- Oct 2020 – Dec 2021 **Joint Ph.D. Program in Mathematics**, *University of Barcelona, Spain*, Funding: China Scholarship Council (CSC)
- Sep 2012 – Jul 2016 **B.Sc. in Mathematics and Biology**, *Nankai University, Tianjin, China*

Publications

Pure Mathematics

- [1] **Indecomposable cardinals** (with Jiachen Yuan), preprint.
- [2] **A new model in which all C-sequences are trivial** (with Assaf Rinot and Jiachen Yuan), *Journal of Symbolic Logic*, published online 2026. DOI: <https://doi.org/10.1017/jsl.2026.10183>
- [3] **Ketonen's question and other cardinal sins** (with Assaf Rinot and Jiachen Yuan), *Israel Journal of Mathematics* (Accepted), preprint: <https://arxiv.org/abs/2408.01547>
- [4] **The vanishing levels of a tree** (with Assaf Rinot and Shira Yadai), *Canadian Journal of Mathematics*, published online 2025. DOI: <https://doi.org/10.4153/S0008414X2510148X>
- [5] **How far is almost strong compactness from strong compactness** (with Jiachen Yuan), *Journal of Mathematical Logic*, 2025. DOI: <https://doi.org/10.1142/S0219061324500090>
- [6] **Full Souslin trees at small cardinals** (with Assaf Rinot and Shira Yadai), *Journal of the London Mathematical Society*, 2024. DOI: <https://doi.org/10.1112/jlms.12957>

- [7] **On the cofinality of the least λ -strongly compact cardinal** (with Jiachen Yuan), *Journal of Symbolic Logic*, 2024. DOI: <https://doi.org/10.1017/jsl.2023.4>

AI Agents

- [8] **D-Mem: A Dual-Process Memory System for LLM Agents** (with Jiachen Yuan and Jason Cai), accepted at the *Second Workshop for Research on Agent Language Models (REALM 2026)* at EMNLP 2026 (archival). arXiv:2603.18631.

Academic Services

Reviewer Archive for Mathematical Logic

Academic Talks

- [1] Indeterminacy of longer Welch game. *Set Theory Seminar in Jerusalem*, Hebrew University of Jerusalem, Israel, Feb 2026.
- [2] Controlling the strength of some large cardinals that carry an indecomposable ultrafilter. *Set Theory Seminar in Jerusalem*, Hebrew University of Jerusalem, Israel, Nov 2025.
- [3] The intersection of a sequence of outer models and sum ultrafilter constructions. *Set Theory Seminar in Jerusalem*, Hebrew University of Jerusalem, Israel, May 2024.
- [4] Intersection of a sequence of outer models. *Bar-Ilan University Set Theory Seminar*, Bar-Ilan University, Israel, May 2024.
- [5] Adding entangled branches to a κ -Suslin tree. *Institute of Mathematics Set Theory Seminar*, Chinese Academy of Sciences, Nov 2023.
- [6] The vanishing level of a tree. *2023 Chinese Annual Conference on Mathematical Logic*, Sun Yat-sen University, Nov 2023.
- [7] Some new constructions of Suslin trees. *Mathematical Logic Seminar*, Wuhan University, Oct 2023.
- [8] How far is almost strong compactness from strong compactness. *Logic Colloquium 2023*, University of Milan, Italy, Jun 2023.
- [9] Identity crisis between δ -strongly compact cardinals. *Tel Aviv University Set Theory Seminar*, Tel Aviv University, Israel, Sep 2022.
- [10] Hierarchies of δ -strongly compact cardinal. *Annual meeting of The Israel Mathematical Union*, Ben Gurion University of the Negev, Israel, Sep 2022.
- [11] A minimal Magidor-type forcing (countable case). *Bar-Ilan University Set Theory Seminar*, Bar-Ilan University, Israel, Sep 2022.
- [12] On the δ -strongly compact cardinal. *Bar-Ilan University Set Theory Seminar*, Bar-Ilan University, Israel, Aug 2022.
- [13] Strong compactness and its variation. *Mathematical Logic Seminar*, Wuhan University, Mar 2022.

- [14] Hierarchies of δ -strongly but not full strongly compact cardinals. *Barcelona Set Theory Seminar*, University of Barcelona, Spain, Sep 2021.
- [15] Hierarchies of δ -strongly but not full strongly compact cardinals. *Logic Workshop in Nanjing*, Nanjing University, Mar 2021.
- [16] Minimal Magidor-type forcing (Countable case). *Prikry Forcing Online*, University of East Anglia, Dec 2020.